

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 15 — Halogen Compounds (Halogenoalkanes)

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	15.1 Reactions (Substitution & Elimination) · 15.2 Mechanisms (SN1 vs SN2) · Hydrolysis Rates · CFCs
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	154 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

15.1 Reactions of Halogenoalkanes

— Nucleophilic substitutions by OH (alcohols), CN (nitriles), and NH₃ (amines); elimination to form alkenes

15.2 Mechanisms of Nucleophilic Substitution

— SN1 (two-step via planar carbocation, racemisation) vs SN2 (concerted backside attack, Walden inversion)

15.2 Hydrolysis Rates & Environmental Chemistry

— Relative rates with aqueous AgNO₃ in ethanol governed by C-X bond enthalpy (C-I > C-Br > C-Cl)

SUBTOPIC 15.2 — MECHANISMS (SN1 & SN2), HYDROLYSIS RATES & CFCS

1 Comparison of SN1 and SN2 Mechanisms — 9701/21/M/J/23/Q7(a)–(d)

HARD · Syllabus 15.2

Halogenoalkanes undergo nucleophilic substitution via either an SN1 or an SN2 pathway depending on the degree of substitution at the halogen-bearing carbon. Fig. 1.1 compares the two mechanisms.

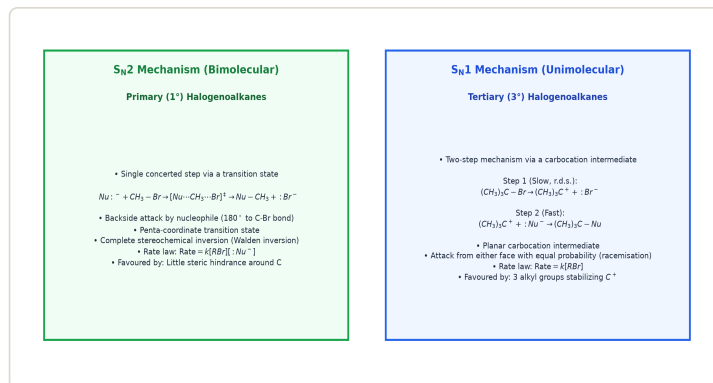


Fig. 1.1: Features and pathways of SN1 and SN2 nucleophilic substitution.

- ((a)) Draw the complete mechanism for the reaction between bromoethane and aqueous sodium hydroxide. Include all partial charges, lone pairs, curly arrows, and the structure of the transition state. [4]

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- ((b)) Draw the complete two-step mechanism for the hydrolysis of 2-bromo-2-methylpropane with aqueous sodium hydroxide, showing the carbocation intermediate. [3]

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- ((c)) Explain why 2-bromo-2-methylpropane reacts predominantly via the SN1 mechanism rather than the SN2 mechanism. [2]

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- ((d)) State and explain the stereochemical outcome when a single optical isomer of a chiral tertiary halogenoalkane undergoes hydrolysis by the SN1 mechanism. [2]

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2 Kinetics and Rate Laws of SN1 vs SN2 — 9701/22/O/N/22/Q7(a)–(b)

HARD · Syllabus 15.2

The rate equations for the alkaline hydrolysis of 1-bromobutane and 2-bromo-2-methylpropane differ fundamentally.

- ((a)) State the rate equation for the hydrolysis of 1-bromobutane with aqueous sodium hydroxide, and state the order of reaction with respect to each reactant. [2]

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- ((b))** State the rate equation for the hydrolysis of 2-bromo-2-methylpropane with aqueous sodium hydroxide, and explain why hydroxide ion concentration does NOT affect the reaction rate. **[2]**
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3 Walden Inversion in SN2 Reactions — 9701/23/M/J/22/Q6(a)–(b)

HARD · Syllabus 15.2

When a chiral primary or secondary halogenoalkane undergoes nucleophilic substitution by the SN2 mechanism, complete inversion of stereochemistry occurs.

- ((a))** Explain why an SN2 reaction causes inversion of configuration (Walden inversion) at the carbon atom. **[2]**
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- ((b))** Draw 3D wedge-and-dash diagrams showing the stereochemical inversion when (S)-2-bromobutane reacts with hydroxide ions via the SN2 pathway. **[2]**
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4 Identifying Mechanism Type by Substrate Structure — 9701/11/M/J/22/Q25

EASY · Syllabus 15.2

Which halogenoalkane reacts with aqueous sodium hydroxide predominantly via the SN2 mechanism?

- (i)** Select the halogenoalkane: **[1]**
- A. 1-bromobutane
 - B. 2-bromobutane
 - C. 2-bromo-2-methylpropane
 - D. 2-bromo-2-methylbutane
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5 Secondary Halogenoalkanes: Borderline Mechanism — 9701/21/O/N/21/Q7(a)–(b)

HARD · Syllabus 15.2

Secondary halogenoalkanes such as 2-bromobutane can undergo nucleophilic substitution via both SN1 and SN2 pathways simultaneously.

- ((a))** Explain why secondary halogenoalkanes can react by both mechanisms. **[2]**
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- ((b))** Suggest how the reaction solvent can be modified to favour the SN1 pathway over the SN2 pathway. **[2]**
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6 Energy Profile of SN1 vs SN2 Reactions — 9701/22/F/M/21/Q7(c)

HARD · Syllabus 15.2

The progress of nucleophilic substitution reactions can be represented using reaction profile diagrams.

- ((a))** Describe the number of energy transition states and intermediate energy troughs present in the reaction profile of an SN1 reaction. [2]

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- ((b))** Describe the reaction profile of an SN2 reaction. [1]

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7 Solvent Effects on Nucleophilic Substitution — 9701/12/M/J/23/Q24

HARD · Syllabus 15.2

Polar aprotic solvents (such as propanone or DMSO) favour SN2 reactions, whereas polar protic solvents favour SN1 reactions.

- (i)** Why do polar aprotic solvents accelerate SN2 reactions? [1]
- A. They form strong hydrogen bonds to the nucleophile, lowering its energy.
 - B. They solvate metal cations strongly but leave nucleophilic anions (e.g. OH, CN) unsolvated and 'naked', making them more reactive.
 - C. They convert halogenoalkanes into carbocations.
 - D. They act as catalysts by donating protons.

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8 Leaving Group Ability in Nucleophilic Substitution — 9701/23/O/N/20/Q5(a)

EASY · Syllabus 15.2

The rate of nucleophilic substitution depends strongly on the nature of the leaving group.

- (i)** List the halide ions in order of increasing leaving group ability (worst leaving group to best leaving group) and justify your answer using bond enthalpy. [3]

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9 Structure of the Transition State in SN2 Reactions — 9701/11/O/N/22/Q22

HARD · Syllabus 15.2

In the transition state of an SN2 reaction between chloromethane and hydroxide ion, [HO...CH3...Cl]:

- (i)** What is the arrangement of the three hydrogen atoms around the central carbon atom? [1]
- A. Tetrahedral
 - B. Pyramidal
 - C. Planar (trigonal planar)
 - D. Linear

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10 Effect of Alkyl Group Size on SN2 Rate — 9701/21/M/J/20/Q4(b)

HARD · Syllabus 15.2

The relative rates of SN2 substitution for 1-halobutane, 1-halo-2-methylpropane, and 1-halo-2,2-dimethylpropane decrease dramatically.

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Explain why steric hindrance near the alpha-carbon severely retards the rate of an S_N2 reaction.

[2]

11

MCQ on Carbocation Stability in S_N1 — 9701/13/M/J/22/Q22

EASY · Syllabus 15.2

Which carbocation formed during the S_N1 hydrolysis of a bromoalkane is the most stable?

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Select the most stable carbocation:

- A. $\text{CH}_3\text{-CH}_2$
- B. $(\text{CH}_3)_2\text{CH}$
- C. $(\text{CH}_3)_3\text{C}$
- D. CH_3

[1]

12

Deducing Mechanism from Experimental Observations — 9701/22/O/N/23/Q7(b)

HARD · Syllabus 15.2

A chemist investigates the hydrolysis of an optically active halogenoalkane R.

((a))

When pure (R)-2-iodobutane is reacted with aqueous NaOH, the product butan-2-ol rotates plane-polarised light in the opposite direction to the starting material. Deduce the primary mechanism operating.

[1]

((b))

Explain how this optical rotation result proves that substitution did NOT occur exclusively via a carbocation intermediate.

[2]

13

Reactivity of Halogenoarenes vs Halogenoalkanes — 9701/12/F/M/22/Q22

HARD · Syllabus 15.2

Chlorobenzene does NOT undergo nucleophilic substitution when boiled with aqueous sodium hydroxide under reflux, whereas 1-chlorobutane hydrolyses smoothly.

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Why is chlorobenzene inert to nucleophilic substitution under normal laboratory conditions?

- A. The chlorine atom is too small to be replaced.
- B. A lone pair of electrons on the chlorine atom overlaps with the delocalised pi system of the benzene ring, strengthening the C-Cl bond with p
- C. Benzene rings attract nucleophiles strongly.
- D. Chlorobenzene is completely insoluble in water.

[1]

SUBTOPIC 15.1 — REACTIONS OF HALOGENOALKANES (SUBSTITUTION & ELIMINATION)

14

Substitution vs Elimination in Halogenoalkanes — 9701/22/M/J/23/Q7(a)-(d)

HARD · Syllabus 15.1

When 2-bromopropane is treated with potassium hydroxide, the reaction pathway is dictated by the choice of solvent and temperature. Fig. 14.1 illustrates the two pathways.

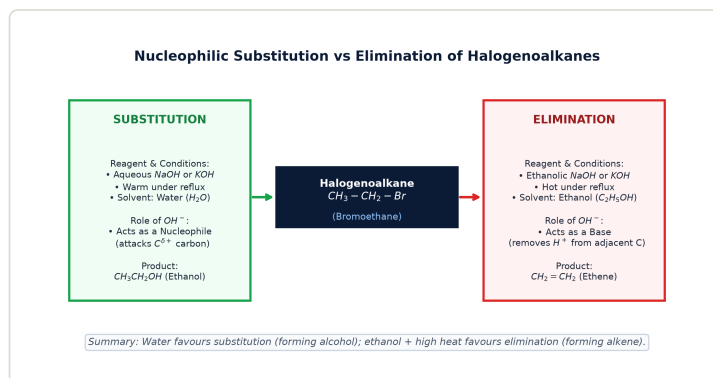


Fig. 14.1: Competition between nucleophilic substitution and elimination for 2-bromopropane.

- ((a)) State the reagent and conditions required to convert 2-bromopropane into propan-2-ol, and identify the role of the hydroxide ion. [2]

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- ((b)) State the reagent and conditions required to convert 2-bromopropane into propene, and identify the role of the hydroxide ion. [2]

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- ((c)) Write the balanced chemical equation for the elimination reaction in (b). [1]

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- ((d)) Draw the complete mechanism for the elimination reaction of 2-bromopropane with hydroxide ions, including all curly arrows. [3]

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15 Reaction of Halogenoalkanes with Ethanolic Potassium Cyanide — 9701/21/O/N/22/Q7(c)-(e)

EASY · Syllabus 15.1

Halogenoalkanes react with potassium cyanide to form nitriles, an important reaction for extending the carbon chain.

- ((a)) State the reagent and essential condition for the reaction of 1-chloropropane with potassium cyanide. [2]

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- ((b)) Write the chemical equation for the reaction of 1-chloropropane with KCN, and give the systematic IUPAC name of the organic product. [2]

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- ((c)) State the formula of the carboxylic acid formed when this nitrile product is refluxed with dilute hydrochloric acid. [1]
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16 Reaction of Halogenoalkanes with Ammonia: Amine Formation — 9701/23/M/J/22/Q6(c)–(e)

HARD · Syllabus 15.1

Bromoethane reacts with ammonia to form ethylamine.

- ((a)) State the conditions required to prepare ethylamine from bromoethane. [2]
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- ((b)) Write the overall balanced equation for the formation of ethylamine from bromoethane and ammonia. [2]
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- ((c)) Explain why an excess of concentrated ethanolic ammonia is used in this preparation. [2]
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17 Isomeric Alkenes from Elimination of 2-Bromobutane — 9701/21/M/J/22/Q7(a)–(c)

HARD · Syllabus 15.1

When 2-bromobutane is heated under reflux with concentrated ethanolic potassium hydroxide, a mixture of three isomeric alkenes is formed.

- ((a)) Draw the displayed formula and state the systematic IUPAC name of the minor alkene product formed by elimination of a hydrogen atom from carbon-1. [2]
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- ((b)) Elimination of a hydrogen atom from carbon-3 produces two stereoisomers. Draw the displayed formulae of both stereoisomers and name them using the E/Z system. [3]
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- ((c)) State which alkene is formed in the greatest yield and suggest why. [1]
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18 Identifying Functional Group Transitions — 9701/11/M/J/23/Q24

EASY · Syllabus 15.1

Which reaction sequence converts 1-bromopropane into butanoic acid in two laboratory steps?

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Select the correct two-step sequence:

- A. Step 1: Aqueous NaOH; Step 2: Acidified K₂Cr₂O₇ reflux
- B. Step 1: Ethanolic KCN reflux; Step 2: Dilute aqueous HCl reflux
- C. Step 1: Ethanolic NH₃; Step 2: Nitrous acid
- D. Step 1: Ethanolic KOH reflux; Step 2: Steam with H₃PO₄

[1]

19

Role of the Hydroxide Ion in Halogenoalkane Reactions — 9701/12/O/N/21/Q23

EASY · Syllabus 15.1

Consider the two reactions:

Reaction 1: CH₃CH₂Br + OH(aq) → CH₃CH₂OH + Br

Reaction 2: CH₃CH₂Br + OH(ethanol) → CH₂=CH₂ + H₂O + Br

0

In which reaction is the hydroxide ion acting as a Brønsted-Lowry base?

- A. Reaction 1 only
- B. Reaction 2 only
- C. Both Reaction 1 and Reaction 2
- D. Neither reaction

[1]

20

Synthesis of Amines from Halogenoalkanes — 9701/22/F/M/23/Q7(a)-(b)

EASY · Syllabus 15.1

Butylamine, CH₃(CH₂)₃NH₂, is prepared from 1-chlorobutane.

((a))

Write the balanced chemical equation for the reaction of 1-chlorobutane with excess ammonia.

[2]

((b))

Explain why butylamine is a basic substance in aqueous solution, referring to its structure.

[2]

21

Elimination of 2-Bromo-2-methylbutane — 9701/21/O/N/23/Q7(b)

HARD · Syllabus 15.1

Heating 2-bromo-2-methylbutane with hot ethanolic KOH produces two isomeric alkenes.

((a))

Draw the skeletal formulae and state the systematic IUPAC names of both alkenes.

[3]

((b))

State which of these two alkenes is formed as the major product, and justify your answer.

[2]

22 Cyanide Nucleophile Structure — 9701/13/O/N/21/Q22

EASY · Syllabus 15.1

In the reaction of a halogenoalkane with potassium cyanide, which atom of the cyanide ion, $[:C\equiv N:]^-$, attacks the carbon atom bearing the halogen?

0

Select the attacking atom and the reason:

- A. Nitrogen, because nitrogen is more electronegative than carbon.
- B. Carbon, because the negative formal charge and more nucleophilic lone pair reside on carbon.
- C. Both atoms attack simultaneously to form a cyclic intermediate.
- D. Potassium, which donates an electron.

[1]

23 Distinguishing Elimination from Substitution by Bromine Test — 9701/22/M/J/21/Q7(b)

EASY · Syllabus 15.1

A student carries out a reaction on 1-bromobutane and collects the organic product.

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Describe a simple chemical test to confirm whether the reaction product was formed by elimination rather than by substitution.

[2]

24 Multi-Step Synthesis: Halogenoalkane to Amide — 9701/23/M/J/20/Q5(c)

HARD · Syllabus 15.1

Outline a two-step reaction scheme to convert bromoethane, CH_3CH_2Br , into propanamide, $CH_3CH_2CONH_2$.

((a))

Identify the reagent, conditions, and intermediate formed in Step 1.

[2]

((b))

Identify the reagent and condition for Step 2.

[1]

SUBTOPIC 15.2 — MECHANISMS (S_N1 & S_N2), HYDROLYSIS RATES & CFCS

25 Experimental Determination of Relative Rates of Hydrolysis — 9701/21/M/J/23/Q7(e)–(g)

HARD · Syllabus 15.2

The relative rates of hydrolysis of 1-chlorobutane, 1-bromobutane, and 1-iodobutane can be compared experimentally using aqueous silver nitrate in ethanol at 50 °C. Fig. 25.1 shows the relative rates and bond enthalpies.

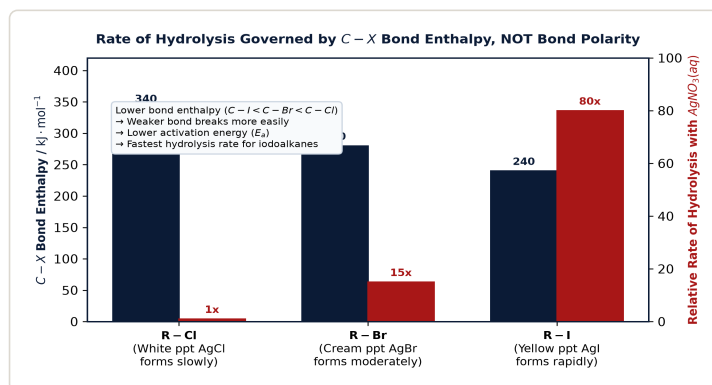


Fig. 25.1: Rates of hydrolysis compared with C-X bond enthalpies.

((a)) Explain why ethanol is added to the reaction mixture along with aqueous silver nitrate. [1]

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((b)) State the observation and the relative time taken for a precipitate to appear for each of the three halogenoalkanes. [3]

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((c)) Write the ionic equation for the formation of the precipitate in the case of 1-iodobutane. [1]

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((d)) Explain the trend in the rates of hydrolysis in terms of bond enthalpies and bond polarities. [3]

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26 Hydrolysis Rates: Primary vs Secondary vs Tertiary — 9701/22/O/N/22/Q7(f)

HARD · Syllabus 15.2

Equal amounts of 1-chlorobutane, 2-chlorobutane, and 2-chloro-2-methylpropane are reacted separately with aqueous silver nitrate in ethanol at 50 °C.

((a)) State the order of reactivity of these three chloroalkanes from slowest to fastest. [1]

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((b)) Explain why 2-chloro-2-methylpropane hydrolyses significantly faster than 1-chlorobutane. [3]

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27 Unreactivity of Fluoroalkanes towards Hydrolysis — 9701/11/M/J/22/Q26

EASY · Syllabus 15.2

When 1-fluorobutane is heated with aqueous silver nitrate in ethanol, no precipitate is observed even after several hours.

() Why does 1-fluorobutane fail to produce a precipitate? [1]

- A. Silver fluoride is soluble in water, and the C-F bond enthalpy is exceptionally high (467 kJ mol⁻¹).
- B. Fluorine is too electronegative to form a covalent bond with carbon.
- C. Fluoride ions reduce silver ions to metallic silver.
- D. 1-fluorobutane evaporates before it can react.

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28

Environmental Impact of Chlorofluorocarbons (CFCs) — 9701/21/O/N/21/Q7(c)-(e)

HARD · Syllabus 15.2

Chlorofluorocarbons (CFCs) such as dichlorodifluoromethane, CCl_2F_2 , were historically used as aerosol propellants and refrigerants. Fig. 28.1 depicts their stratospheric photolysis.

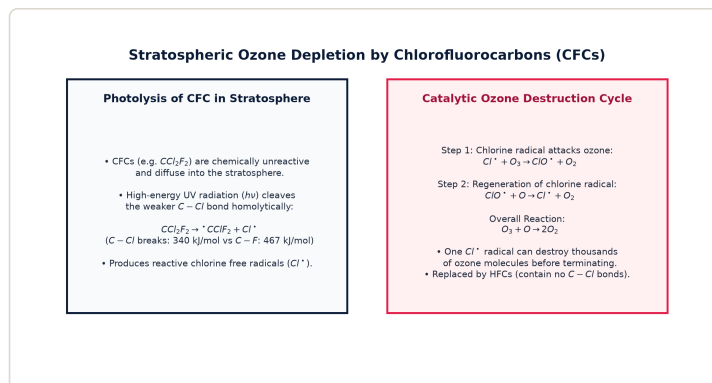


Fig. 28.1: Photodissociation of CFCs and the catalytic ozone destruction cycle.

((a)) Explain why CFCs are chemically stable and inert in the lower atmosphere (troposphere). **[1]**

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((b)) When CFC molecules diffuse into the stratosphere, high-energy UV radiation causes bond fission. Write the equation for the photolysis of CCl_2F_2 and explain why the C–Cl bond breaks rather than the C–F bond. **[2]**

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((c)) Write two equations that show how chlorine radicals catalytically destroy stratospheric ozone, and construct the overall equation. **[3]**

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((d)) Name a modern class of replacement compounds for CFCs that have zero ozone depletion potential, and explain why they do not destroy ozone. **[2]**

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29

Precipitate Solubility in Aqueous Ammonia Test — 9701/22/M/J/22/Q7(b)

EASY · Syllabus 15.2

The silver halide precipitates obtained from halogenoalkane hydrolysis can be confirmed using aqueous ammonia.

0 Complete the following table of solubilities for AgCl, AgBr, and AgI in dilute and concentrated aqueous ammonia. **[3]**

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30 Kinetics of Hydrolysis with Different Leaving Groups — 9701/12/M/J/21/Q24

EASY · Syllabus 15.2

Which factor primarily accounts for the observation that 1-iodobutane hydrolyses faster than 1-chlorobutane?

0

Select the primary factor:

- A. The C-I bond is more polar than the C-Cl bond.
- B. The C-I bond enthalpy is lower than the C-Cl bond enthalpy.
- C. Iodine is more electronegative than chlorine.
- D. Iodide ions are smaller than chloride ions.

[1]

31 Atmospheric Lifetime and Greenhouse Impact of HFCs — 9701/23/O/N/21/Q6(c)

HARD · Syllabus 15.2

Although hydrofluorocarbons (HFCs) do not deplete stratospheric ozone, their widespread use has raised new environmental concerns.

((a))

State the major global environmental concern associated with HFC emissions into the atmosphere.

[1]

((b))

Explain why HFC molecules absorb infrared radiation so effectively.

[2]

32 Deducing Halogen Identity by Gravimetric Precipitation — 9701/21/M/J/21/Q7(c)

HARD · Syllabus 15.2

A 0.685 g sample of a 1-haloalkane, C_4H_9X , was completely hydrolysed with excess aqueous sodium hydroxide. The solution was acidified with dilute HNO_3 and treated with excess $AgNO_3(aq)$. The dried precipitate weighed 0.940 g. (Ar: C = 12.0, H = 1.0, Ag = 107.9; Cl = 35.5, Br = 79.9, I = 126.9)

((a))

Calculate the number of moles of C_4H_9X in the sample if X is bromine (M_r of C_4H_9Br = 137.0).

[1]

((b))

Determine whether the precipitate is $AgCl$, $AgBr$, or AgI by calculating the expected mass of precipitate for each.

[3]

SUBTOPIC 15.1 — REACTIONS OF HALOGENOALKANES (SUBSTITUTION & ELIMINATION)

33 Comparison of Halogenoalkanes with Hydroxyalkanes — 9701/13/O/N/22/Q23

EASY · Syllabus 15.1

1-bromobutane boils at 102 °C, whereas butan-1-ol boils at 118 °C, despite 1-bromobutane having a much higher relative molecular mass (M_r : 137.0 vs 74.0).

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Why is the boiling point of butan-1-ol higher than that of 1-bromobutane?

- A. 1-bromobutane molecules are spherical.
- B. Butan-1-ol molecules form intermolecular hydrogen bonds, which are stronger than the permanent dipole-dipole and London dispersion forces.
- C. The C-Br bond is non-polar.
- D. Butan-1-ol has a higher density.

[1]

SUBTOPIC 15.2 — MECHANISMS (SN1 & SN2), HYDROLYSIS RATES & CFCS

34 Mechanism of Alkyl Halide Solvolysis — 9701/22/F/M/20/Q4(b)

HARD · Syllabus 15.2

Solvolysis is a nucleophilic substitution in which the solvent molecule acts as the attacking nucleophile.

((a)) Write the equation for the solvolysis of 2-iodo-2-methylpropane in pure water, showing the intermediate oxonium ion. [2]

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((b)) Explain why solvolysis occurs rapidly in water but very slowly in ethanol. [2]

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35 MCQ on Ozone Depletion Chemistry — 9701/11/F/M/23/Q23

EASY · Syllabus 15.2

In the stratosphere, what role is played by the chlorine radical, Cl^* , in the decomposition of ozone?

0 Select the correct role: [1]

- A. Initiator consumed in the reaction
- B. Homogeneous catalyst
- C. Heterogeneous catalyst
- D. Reducing agent oxidised to chlorate

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SUBTOPIC 15.1 — REACTIONS OF HALOGENOALKANES (SUBSTITUTION & ELIMINATION)

36 Preparation of Halogenoalkanes from Alcohols — 9701/21/O/N/20/Q6(b)

HARD · Syllabus 15.1

Halogenoalkanes can be prepared from alcohols using various halogenating reagents.

((a)) Write the balanced equation for the reaction of propan-1-ol with phosphorus pentachloride, PCl_5 , at room temperature, and state two observations. [3]

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((b)) Explain why thionyl chloride, SOCl_2 , is preferred over PCl_5 for the synthesis of chloroalkanes in organic laboratories. [2]

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SUBTOPIC 15.2 — MECHANISMS (SN1 & SN2), HYDROLYSIS RATES & CFCS

37 Identifying Hydrolysis Products by Mass Spectrometry — 9701/22/M/J/21/Q7(d)

HARD · Syllabus 15.2

The mass spectrum of a bromoalkane exhibits two molecular ion peaks of almost equal intensity at $m/z = 136$ and $m/z = 138$.

((a)) Explain why two molecular ion peaks of equal intensity are observed. **[2]**

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((b)) Deduce the molecular formula of this bromoalkane. (Ar: C = 12.0, H = 1.0; Br = 79, ⁷⁹Br = 81) **[2]**

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SUBTOPIC 15.1 — REACTIONS OF HALOGENOALKANES (SUBSTITUTION & ELIMINATION)

38 MCQ on Halogenoalkane Nucleophilic Attack — 9701/12/M/J/22/Q25

EASY · Syllabus 15.1

Which species CANNOT act as a nucleophile in a substitution reaction with a halogenoalkane?

0 **[1]**

Select the species:

- A. NH₃
- B. H₂O
- C. NH₄⁺
- D. CN⁻

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SUBTOPIC 15.2 — MECHANISMS (S_N1 & S_N2), HYDROLYSIS RATES & CFCs

39 Comparative Rates: Bromoethane vs Iodoethane — 9701/23/M/J/21/Q6(b)

EASY · Syllabus 15.2

Bromoethane and iodoethane are reacted with aqueous sodium hydroxide under identical conditions.

0 State which compound reacts faster and give the single most important factor that accounts for this difference. **[2]**

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40 Montreal Protocol and Replacement of CFCs — 9701/11/M/J/21/Q24

EASY · Syllabus 15.2

The international Montreal Protocol phased out the manufacture and use of chlorofluorocarbons.

0 Which compound is most suitable as an environmentally acceptable propellant in asthma inhalers? **[1]**

- A. CCl₃F
- B. CCl₂F₂
- C. CF₃CH₂F (1,1,1,2-tetrafluoroethane)
- D. CCl₄

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41 Mechanistic Pathway for Hydrolysis of Benzyl Chloride — 9701/21/O/N/23/Q7(d)

HARD · Syllabus 15.2

Benzyl chloride, C₆H₅CH₂Cl, hydrolyses rapidly via an S_N1 pathway even though it is a primary halogenoalkane.

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Explain why the primary benzyl carbocation, $\text{C}_6\text{H}_5\text{CH}_2^+$, is exceptionally stable.

[2]

SUBTOPIC 15.1 — REACTIONS OF HALOGENOALKANES (SUBSTITUTION & ELIMINATION)

42

Identifying Elimination Products from Secondary Haloalkanes — 9701/13/O/N/21/Q23

EASY · Syllabus 15.1

When 3-bromopentane is heated with concentrated ethanolic sodium hydroxide, what is the systematic IUPAC name of the alkene product?

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Select the correct alkene name:

- A. Pent-1-ene
- B. Pent-2-ene
- C. 2-methylbut-2-ene
- D. Cyclopentane

[1]

43

Nucleophilic Substitution with Carboxylate Salts — 9701/22/F/M/22/Q7(b)

HARD · Syllabus 15.1

Halogenoalkanes can react with carboxylate salts to form esters: $\text{R-X} + \text{R}'\text{COO}^- \rightarrow \text{R}'\text{COOR} + \text{X}^-$.

((a))

Write the balanced equation for the reaction of bromoethane with sodium ethanoate.

[1]

((b))

State the systematic IUPAC name of the ester formed.

[1]

44

MCQ on Halogenoalkane Boiling Points — 9701/12/O/N/22/Q24

EASY · Syllabus 15.1

Which halogenoalkane has the highest boiling point?

0

Select the compound with highest boiling point:

- A. 1-chlorobutane
- B. 1-bromobutane
- C. 1-iodobutane
- D. 1-fluorobutane

[1]

SUBTOPIC 15.2 — MECHANISMS ($\text{S}_\text{N}1$ & $\text{S}_\text{N}2$), HYDROLYSIS RATES & CFCS

45

Acid-Catalysed Hydrolysis of Haloalkanes — 9701/21/M/J/22/Q7(e)

HARD · Syllabus 15.2

Explain why adding aqueous silver nitrate acidified with dilute nitric acid accelerates the rate of hydrolysis of halogenoalkanes.

0

State the role of Ag^+ ions in facilitating the cleavage of the C-X bond.

[2]

46 Ozone Destruction Catalytic Cycles with Nitrogen Oxides and Chlorine — 9701/11/O/N/21/Q23

HARD · Syllabus 15.2

Chlorine monoxide radicals, ClO^* , can also react with nitrogen dioxide in the stratosphere: $\text{ClO}^* + \text{NO}_2 \rightarrow \text{ClONO}_2$.

(0)

What is the significance of the formation of chlorine nitrate, ClONO_2 , in stratospheric chemistry?

- A. It acts as an active catalyst that accelerates ozone destruction.
- B. It acts as a temporary 'reservoir molecule' that removes active Cl^* and NO_2^* radicals from the catalytic cycles, temporarily reducing ozone destruction.
- C. It permanently deposits chlorine onto the Earth's surface.
- D. It decomposes directly into ozone.

[1]

47 Synthesis of Deuterated Alcohols via $\text{S}_\text{N}2$ — 9701/22/O/N/20/Q6(b)

HARD · Syllabus 15.2

Bromoethane is hydrolysed using sodium deuterioxide in heavy water, $\text{NaOD} / \text{D}_2\text{O}$.

((a))

Draw the structural formula of the deuterated alcohol formed.

[1]

((b))

State the molecular mass of the deuterated alcohol product. (Ar: C = 12.0, H = 1.0, D = 2.0, O = 16.0)

[1]

SUBTOPIC 15.1 — REACTIONS OF HALOGENOALKANES (SUBSTITUTION & ELIMINATION)

48 MCQ on Halogenoalkane Nucleophilic Attack — 9701/13/M/J/23/Q25

EASY · Syllabus 15.1

Which reaction condition specifically favours elimination over substitution when reacting a halogenoalkane with potassium hydroxide?

(0)

Select the favoured condition:

- A. Low temperature in dilute aqueous solution
- B. High temperature in concentrated ethanolic solution
- C. Room temperature in water-ethanol (50:50) mixture
- D. High pressure with liquid ammonia

[1]

SUBTOPIC 15.2 — MECHANISMS ($\text{S}_\text{N}1$ & $\text{S}_\text{N}2$), HYDROLYSIS RATES & CFCS

49 Qualitative Analysis of an Unknown Halogenoalkane — 9701/22/M/J/23/Q7(f)

EASY · Syllabus 15.2

An unknown halogenoalkane Q is warmed with aqueous sodium hydroxide. The mixture is cooled, acidified with dilute nitric acid, and treated with aqueous silver nitrate. A cream precipitate forms that is insoluble in dilute aqueous ammonia but dissolves in concentrated aqueous ammonia.

((a))

Identify the halogen present in compound Q.

[1]

((b))

Write the ionic equation for the formation of the precipitate.

[1]

SUBTOPIC 15.1 — REACTIONS OF HALOGENOALKANES (SUBSTITUTION & ELIMINATION)

50 Synoptic Reaction Scheme Involving Halogenoalkanes — 9701/21/O/N/23/Q7(a)–(f)

HARD · Syllabus 15.1

Consider the synthetic pathways starting from 1-bromobutane:

- Pathway 1: 1-bromobutane + Reagent A → Butan-1-ol
- Pathway 2: 1-bromobutane + Reagent B → Compound C (C₅H₉N)
- Pathway 3: 1-bromobutane + Reagent D → But-1-ene

((a)) Identify Reagents A, B, and D, and state the essential conditions for each pathway. [3]

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((b)) Give the systematic IUPAC name of Compound C. [1]

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((c)) State the reagent and conditions required to convert Compound C into pentanoic acid. [2]

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((d)) State the reagent and conditions required to convert Compound C into pentan-1-amine. [2]

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Mark Scheme — Topic 15 — Halogen Compounds (Halogenoalkanes)

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	Curly arrow from lone pair on :OH to C(δ^+) of CH ₃ CH ₂ -Br [1] Curly arrow showing heterolytic cleavage of C-Br bond to Br(δ^-) [1] Penta-coordinate transition state drawn with partial bonds to OH and Br and overall negative charge [1] Inverted product CH ₃ CH ₂ OH and :Br drawn correctly [1]	4
1(b)	Step 1: Curly arrow showing heterolytic fission of C-Br bond in (CH ₃) ₃ C-Br to form (CH ₃) ₃ C and :Br [1] Structure of planar tertiary carbocation intermediate (CH ₃) ₃ C [1] Step 2: Curly arrow from lone pair on :OH to the C of the carbocation to form (CH ₃) ₃ COH [1]	3
1(c)	The three bulky methyl groups create severe steric hindrance, preventing backside attack by the nucleophile in S _N 2 [1] The three electron-donating methyl groups stabilize the tertiary carbocation intermediate via the positive inductive effect, facilitating the S _N 1 pathway [1]	2
1(d)	A racemic mixture (optically inactive) is formed [1] The carbocation intermediate is planar at the C centre; the nucleophile has an equal probability of attacking from either the front face or the rear face, giving equimolar amounts of both enantiomers [1]	2
2(a)	Rate = k [CH ₃ CH ₂ CH ₂ CH ₂ Br] [OH ⁻] [1] First order with respect to 1-bromobutane and first order with respect to hydroxide ions (second order overall) [1]	2
2(b)	Rate = k [(CH ₃) ₃ CBr] [1] The rate-determining step is the unimolecular heterolytic ionization of the C-Br bond; hydroxide ions only participate in the subsequent fast second step [1]	2
3(a)	The nucleophile must attack the central carbon atom from the side directly opposite (180°) to the leaving bromide group [1] As the new C-Nu bond forms and the C-Br bond breaks, the remaining three bonds are pushed through like an umbrella inverting in a gust of wind [1]	2
3(b)	Starting reactant drawn with correct 3D tetrahedral geometry around chiral carbon [1] Product (R)-butan-2-ol drawn showing complete inversion of the spatial configuration around the chiral carbon [1]	2
4	Correct Answer: A [1] Explanation: 1-bromobutane is a primary (1°) halogenoalkane with minimal steric hindrance around the alpha carbon, allowing unhindered backside attack by the nucleophile via the S _N 2 mechanism.	1
5(a)	The two alkyl groups provide intermediate steric hindrance (allowing backside attack for S _N 2) [1] and also provide moderate positive inductive stabilization to a secondary carbocation intermediate (allowing S _N 1) [1]	2
5(b)	Use a polar protic solvent (such as water or aqueous ethanol) [1] Polar protic solvents solvate and stabilise the leaving halide anion and carbocation intermediate, significantly lowering the activation energy for the S _N 1 step [1]	2
6(a)	Two activation energy peaks (two transition states) [1] One shallow energy minimum (trough) between them representing the carbocation intermediate [1]	2
6(b)	A single activation energy peak (one transition state) with no reaction intermediate [1]	1
7	Correct Answer: B [1] Explanation: Polar aprotic solvents solvate cations effectively via dipole interactions but cannot form hydrogen bonds to anions, leaving nucleophilic anions relatively unshielded ('naked') and with higher nucleophilicity for S _N 2 backside attack.	1
8	Order: F < Cl < Br < I [1] The C-I bond has the lowest bond enthalpy (240 kJ mol ⁻¹), making it easiest to cleave [1] The C-F bond has the highest bond enthalpy (467 kJ mol ⁻¹), making fluoride an exceptionally poor leaving group [1]	3
9	Correct Answer: C [1] Explanation: In the penta-coordinate transition state of an S _N 2 reaction, the central carbon is sp ² hybridised; the three non-reacting groups (the three H atoms) lie in a trigonal planar arrangement at 120° angles, while the incoming nucleophile and leaving group occupy axial positions at 180°.	1
10	Bulky alkyl groups on the adjacent beta-carbon crowd the rear face of the alpha-carbon [1] This physically shields the C(δ^+) atom from the trajectory of the approaching nucleophile, increasing the activation energy of the transition state [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
11	Correct Answer: C [1] Explanation: (CH ₃) ₃ C is a tertiary carbocation with three electron-donating methyl groups that disperse the positive charge through positive inductive effects, making it the most stable.	1
12(a)	S _N 2 mechanism [1]	1
12(b)	An S _N 1 mechanism proceeds via a planar carbocation intermediate which yields a 50:50 racemic mixture with ZERO net optical rotation [1] Retention of net optical activity (with inverted sign) confirms inversion of configuration characteristic of backside attack in the S _N 2 mechanism [1]	2
13	Correct Answer: B [1] Explanation: The lone pair on chlorine delocalises into the aromatic pi electron system, shortening and strengthening the C–Cl bond (partial double bond character, higher bond enthalpy). Additionally, the electron-rich pi cloud repels approaching nucleophiles.	1
14(a)	Reagents/conditions: Aqueous sodium hydroxide (or KOH), warm under reflux [1] Role of OH: Acts as a nucleophile (electron pair donor) [1]	2
14(b)	Reagents/conditions: Ethanolic sodium hydroxide (or KOH in ethanol), hot under reflux [1] Role of OH: Acts as a Brønsted–Lowry base (proton acceptor) [1]	2
14(c)	CH ₃ –CH(Br)–CH ₃ + OH → CH ₃ –CH=CH ₂ + H ₂ O + Br (or with KOH → KBr + H ₂ O) [1]	1
14(d)	Curly arrow from lone pair on :OH to a hydrogen atom on an adjacent methyl carbon (C1 or C3) [1] Curly arrow from the C–H bond to form the C=C pi bond [1] Curly arrow from the C–Br bond to the bromine atom showing loss of :Br leaving group [1]	3
15(a)	Reagent: Potassium cyanide (KCN) dissolved in ethanol (ethanolic KCN) [1] Condition: Heat under reflux [1]	2
15(b)	CH ₃ CH ₂ CH ₂ Cl + KCN → CH ₃ CH ₂ CH ₂ CN + KCl (or with CN) [1] Name: Butanenitrile [1]	2
15(c)	CH ₃ CH ₂ CH ₂ COOH (butanoic acid) [1]	1
16(a)	Ethanolic ammonia in a sealed tube (under pressure) / heated under pressure [1] Excess ammonia [1]	2
16(b)	CH ₃ CH ₂ Br + 2NH ₃ → CH ₃ CH ₂ NH ₂ + NH ₄ Br [2] (1 mark for species, 1 mark for balancing with 2 NH ₃)	2
16(c)	Ethylamine contains a lone pair on nitrogen and acts as a nucleophile [1] Using excess ammonia prevents further substitution (poly-alkylation) to secondary amines (diethylamine), tertiary amines, or quaternary ammonium salts [1]	2
17(a)	Displayed formula of but-1-ene: CH ₂ =CH–CH ₂ –CH ₃ [1] Name: But-1-ene [1]	2
17(b)	Displayed formula of (Z)-but-2-ene (cis-but-2-ene) [1] Displayed formula of (E)-but-2-ene (trans-but-2-ene) [1] Correctly named as (E)-but-2-ene and (Z)-but-2-ene [1]	3
17(c)	(E)-but-2-ene (trans-but-2-ene) is the major product because it is the most thermodynamically stable alkene (less steric hindrance between methyl groups) [1]	1
18	Correct Answer: B [1] Explanation: Step 1 with ethanolic KCN extends the carbon chain from 3 to 4 carbons to form butanenitrile (CH ₃ CH ₂ CH ₂ CN). Step 2 hydrolyses the nitrile under reflux with dilute acid to yield butanoic acid.	1
19	Correct Answer: B [1] Explanation: In Reaction 2 (elimination), OH accepts a proton (H) from the adjacent carbon, acting as a Brønsted–Lowry base. In Reaction 1, OH donates an electron pair to carbon, acting as a nucleophile.	1
20(a)	CH ₃ (CH ₂) ₃ Cl + 2NH ₃ → CH ₃ (CH ₂) ₃ NH ₂ + NH ₄ Cl [2] (1 mark for species, 1 mark for balancing)	2
20(b)	The nitrogen atom possesses a non-bonding lone pair of electrons [1] It can accept a proton (H) from water molecules: RNH ₂ + H ₂ O ⇌ RNH ₃ + OH [1]	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
21(a)	Alkene 1: 2-methylbut-2-ene, $(\text{CH}_3)_2\text{C}=\text{CHCH}_3$ [1] Alkene 2: 2-methylbut-1-ene, $\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3$ [1] Both skeletal structures drawn correctly [1]	3
21(b)	Major product: 2-methylbut-2-ene [1] The double bond is more highly substituted (trisubstituted vs disubstituted), making it thermodynamically more stable [1]	2
22	Correct Answer: B [1] Explanation: The formal negative charge in the cyanide ion resides on the carbon atom: $[\text{:C}\equiv\text{N:}]^-$. Carbon donates its lone pair to form a stable C-C bond (bond enthalpy 347 kJ mol ⁻¹ vs C-N 305 kJ mol ⁻¹).	1
23	Add bromine water (or aqueous KMnO_4) to the product [1] Elimination product (butene) will rapidly decolourise bromine water from orange-brown to colourless; substitution product (butanol) will not decolourise bromine water [1]	2
24(a)	Reagents/conditions: Ethanolic potassium cyanide (KCN), heat under reflux [1] Intermediate: Propanenitrile, $\text{CH}_3\text{CH}_2\text{CN}$ [1]	2
24(b)	Partial hydrolysis using moderately concentrated acid (or alkaline H_2O_2) to form propanamide [1]	1
25(a)	Ethanol acts as a common mutual solvent to dissolve both the halogenoalkane (covalent organic) and aqueous silver nitrate (ionic water solution), preventing two immiscible layers [1]	1
25(b)	1-iodobutane: Yellow precipitate of AgI forms almost immediately (fastest) [1] 1-bromobutane: Cream precipitate of AgBr forms after a short delay / moderate rate [1] 1-chlorobutane: White precipitate of AgCl forms very slowly / after prolonged heating [1]	3
25(c)	$\text{Ag}(\text{aq}) + \text{I}(\text{aq}) \rightarrow \text{AgI}(\text{s})$ [1]	1
25(d)	Rate increases in the order: $\text{R-Cl} < \text{R-Br} < \text{R-I}$ [1] Although the C-Cl bond is the most polar (greatest δ^+ on carbon), bond enthalpy is the dominant factor determining reactivity [1] C-I has the lowest bond enthalpy (240 kJ mol ⁻¹ vs C-Cl 340 kJ mol ⁻¹); the weaker bond requires less energy to cleave, leading to a lower activation energy and faster rate [1]	3
26(a)	1-chlorobutane (primary) < 2-chlorobutane (secondary) < 2-chloro-2-methylpropane (tertiary) [1]	1
26(b)	2-chloro-2-methylpropane reacts via the $\text{S}_{\text{N}}1$ mechanism involving a tertiary carbocation intermediate [1] The tertiary carbocation, $(\text{CH}_3)_3\text{C}^+$, is strongly stabilized by the positive inductive electron donation from three methyl groups [1] This results in a much lower activation energy than the concerted $\text{S}_{\text{N}}2$ pathway for 1-chlorobutane [1]	3
27	Correct Answer: A [1] Explanation: The C-F bond is extremely strong (467 kJ mol ⁻¹), making cleavage by water/OH impractically slow at 50 °C. Furthermore, silver fluoride (AgF) is freely soluble in water, so even if trace hydrolysis occurred, no precipitate would form.	1
28(a)	CFCs contain only strong, unreactive C-Cl and C-F bonds with high bond enthalpies and low polarity that resist attack by atmospheric oxidizing agents [1]	1
28(b)	Equation: $\text{CCl}_2\text{F}_2 + \text{UV} \rightarrow \cdot\text{CClF}_2 + \text{Cl}\cdot$ [1] The C-Cl bond enthalpy (340 kJ mol ⁻¹) is significantly lower than the C-F bond enthalpy (467 kJ mol ⁻¹), so the C-Cl bond breaks homolytically under UV light [1]	2
28(c)	Step 1: $\text{Cl}\cdot + \text{O}_3 \rightarrow \text{ClO}\cdot + \text{O}_2$ [1] Step 2: $\text{ClO}\cdot + \text{O} \rightarrow \text{Cl}\cdot + \text{O}_2$ [1] Overall: $\text{O}_3 + \text{O} \rightarrow 2\text{O}_2$ [1]	3
28(d)	Hydrofluorocarbons (HFCs) [1] HFCs contain only C, H, and F atoms (no chlorine atoms); they cannot generate chlorine free radicals to catalyse ozone breakdown [1]	2
29	AgCl: Dissolves in dilute $\text{NH}_3(\text{aq})$ [1] AgBr: Insoluble in dilute $\text{NH}_3(\text{aq})$, dissolves in concentrated $\text{NH}_3(\text{aq})$ [1] AgI: Insoluble in both dilute and concentrated $\text{NH}_3(\text{aq})$ [1]	3
30	Correct Answer: B [1] Explanation: The C-I bond enthalpy (240 kJ mol ⁻¹) is significantly lower than the C-Cl bond enthalpy (340 kJ mol ⁻¹). Bond enthalpy determines the activation energy required to break the carbon-halogen bond during hydrolysis.	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
31(a)	HFCs are potent greenhouse gases with very high global warming potential (GWP) [1]	1
31(b)	HFCs contain highly polar C-F and C-H bonds with large dipole moments [1] Their bond vibration frequencies fall precisely within the atmospheric infrared window (700 – 1400 cm ⁻¹), efficiently trapping outgoing terrestrial radiation [1]	2
32(a)	Moles of C ₄ H ₉ Br = 0.685 / 137.0 = 0.00500 mol [1]	1
32(b)	If X = Br, moles of AgBr formed = 0.00500 mol; Mr(AgBr) = 107.9 + 79.9 = 187.8 [1] Expected mass = 0.00500 * 187.8 = 0.939 g [1] 0.939 g matches the experimental mass of 0.940 g; therefore, halogen X is bromine (AgBr) [1]	3
33	Correct Answer: B [1] Explanation: Butan-1-ol contains a polar –O–H group capable of forming intermolecular hydrogen bonds between molecules. Hydrogen bonds are significantly stronger than the dipole-dipole and London forces between 1-bromobutane molecules.	1
34(a)	(CH ₃) ₃ C–I + H ₂ O → (CH ₃) ₃ C–OH ₂ + I [1] (CH ₃) ₃ C–OH ₂ + H ₂ O → (CH ₃) ₃ COH + H ₃ O [1]	2
34(b)	Water has a higher dielectric constant / greater polarity than ethanol [1] Water solvates and stabilizes the carbocation and iodide ion intermediates more effectively, lowering the activation energy for ionization [1]	2
35	Correct Answer: B [1] Explanation: The chlorine radical is in the same gas phase as ozone (homogeneous) and is regenerated at the end of the two-step propagation cycle (catalyst), enabling a single radical to destroy thousands of ozone molecules.	1
36(a)	CH ₃ CH ₂ CH ₂ OH + PCl ₅ → CH ₃ CH ₂ CH ₂ Cl + POCl ₃ + HCl [1] Observations: Steamy acidic fumes of HCl gas [1] Vigorous reaction / bubbling / heat evolved [1]	3
36(b)	CH ₃ CH ₂ CH ₂ OH + SOCl ₂ → CH ₃ CH ₂ CH ₂ Cl + SO ₂ (g) + HCl(g) [1] Both by-products (SO ₂ and HCl) are gases and escape from the reaction mixture, leaving pure chloroalkane without requiring complex distillation separation [1]	2
37(a)	Bromine exists naturally as two isotopes, Br and ⁸¹ Br, in approximately equal 1:1 abundance [1] The peak at m/z = 136 corresponds to the molecule containing Br; the peak at m/z = 138 corresponds to ⁸¹ Br [1]	2
37(b)	Mass of alkyl group R = 136 – 79 = 57 [1] C _n H _{2n+1} = 57 ⇒ 14n + 1 = 57 ⇒ 14n = 56 ⇒ n = 4 ⇒ C ₄ H ₉ Br [1]	2
38	Correct Answer: C [1] Explanation: The ammonium ion (NH ₄ ⁺) has no non-bonding lone pair of electrons (its octet is fully bonded) and carries a positive charge, meaning it cannot donate an electron pair to act as a nucleophile.	1
39	Iodoethane reacts faster [1] The C–I bond enthalpy (240 kJ mol ⁻¹) is lower than the C–Br bond enthalpy (280 kJ mol ⁻¹), so less energy is required to break the bond [1]	2
40	Correct Answer: C [1] Explanation: CF ₃ CH ₂ F (HFA-134a / HFC-134a) contains no chlorine atoms, poses zero risk to stratospheric ozone, and is non-toxic.	1
41	The empty p-orbital on the positively charged CH ₂ carbon overlaps with the adjacent delocalised pi system of the benzene ring [1] The positive charge is delocalised across the aromatic ring (ortho and para positions), significantly stabilizing the intermediate carbocation [1]	2
42	Correct Answer: B [1] Explanation: 3-bromopentane is CH ₃ CH ₂ CH(Br)CH ₂ CH ₃ . Removing H from C2 or C4 and Br from C3 yields pent-2-ene (CH ₃ CH=CHCH ₂ CH ₃) as the sole constitutional product.	1
43(a)	CH ₃ CH ₂ Br + CH ₃ COONa → CH ₃ COOCH ₂ CH ₃ + NaBr [1]	1
43(b)	Ethyl ethanoate [1]	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
44	Correct Answer: C [1] Explanation: 1-iodobutane has the greatest number of electrons (66 electrons), leading to the most polarisable electron cloud and the strongest instantaneous dipole-induced dipole (London dispersion) forces.	1
45	Ag ions coordinate strongly to the halogen atom, polarising and weakening the C-X bond [1] Precipitation of insoluble AgX solid provides a strong thermodynamic driving force, effectively pulling the reaction forward [1]	2
46	Correct Answer: B [1] Explanation: Chlorine nitrate acts as a temporary chemical reservoir; it ties up active chlorine and nitrogen radicals in an inactive form, preventing them from participating in ozone-destroying catalytic cycles until released.	1
47(a)	CH ₃ CH ₂ -OD (or CH ₃ CH ₂ OD) [1]	1
47(b)	Mr = 2(12.0) + 5(1.0) + 16.0 + 2.0 = 47.0 [1]	1
48	Correct Answer: B [1] Explanation: Hot, concentrated ethanolic KOH favours elimination (forming an alkene), whereas warm, dilute aqueous NaOH favours substitution (forming an alcohol).	1
49(a)	Bromine (or bromide, Br) [1]	1
49(b)	Ag(aq) + Br(aq) → AgBr(s) [1]	1
50(a)	Reagent A: Aqueous NaOH (or KOH), warm under reflux [1] Reagent B: Ethanolic KCN, heat under reflux [1] Reagent D: Ethanolic NaOH (or KOH), hot under reflux [1]	3
50(b)	Pentanenitrile [1]	1
50(c)	Dilute hydrochloric acid (or dilute H ₂ SO ₄), heat under reflux (acid hydrolysis) [2]	2
50(d)	LiAlH ₄ in dry ether (or H ₂ with Ni catalyst at elevated temp/pressure) [2]	2